

Internal Projectivity Implies Properness over $\mathbf{Sp}$

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Theorem 1. *Let $\mathcal{C} \in \mathbf{Cat}_{\text{st}}^{\text{dual}}$. If $\underline{\mathbf{Hom}}^{\text{dual}}(\mathcal{C}, -)$ preserves short exact sequences, then $\mathcal{C}$ is proper.*

Proof. We first construct a proper quotient cover. By [1, Proposition 1.17], there are a small stable category $\mathcal{A}$ and a strongly continuous fully faithful functor $j : \mathcal{C} \hookrightarrow \text{Ind}(\mathcal{A})$. Choose a simplicial set presenting $\mathcal{A}$, let K be its category of simplices, and write $\lambda : K \rightarrow \mathcal{A}$ for the last-vertex localization [3, Tag 01NC]. Put

$$\mathcal{P}_{\mathbf{Sp}}(K) := \text{Fun}(K^{\text{op}}, \mathbf{Sp}).$$

Restriction along λ gives a fully faithful functor

$$R : \text{Ind}(\mathcal{A}) = \text{Fun}^{\text{ex}}(\mathcal{A}^{\text{op}}, \mathbf{Sp}) \longrightarrow \mathcal{P}_{\mathbf{Sp}}(K).$$

It preserves colimits, so its left adjoint $L : \mathcal{P}_{\mathbf{Sp}}(K) \rightarrow \text{Ind}(\mathcal{A})$ is a strongly continuous localization. Moreover $\mathcal{P}_{\mathbf{Sp}}(K)$ is proper: its compact objects are generated by the representables h_k , and

$$\text{Map}(h_k, h_l) \simeq \Sigma_+^{\infty} \text{Hom}_K(k, l)$$

is finite. Form the pullback

$$\mathcal{P} := \mathcal{C} \times_{\text{Ind}(\mathcal{A})} \mathcal{P}_{\mathbf{Sp}}(K).$$

By [1, Proposition 1.87(2)], the projection $q : \mathcal{P} \rightarrow \mathcal{C}$ is a strongly continuous localization and $\mathcal{P} \hookrightarrow \mathcal{P}_{\mathbf{Sp}}(K)$ is strongly continuous and fully faithful. Hence $\mathcal{P}$ is proper by [1, Lemma C.8(1)].

Internal projectivity makes

$$\underline{\mathbf{Hom}}^{\text{dual}}(\mathcal{C}, q) : \underline{\mathbf{Hom}}^{\text{dual}}(\mathcal{C}, \mathcal{P}) \rightarrow \underline{\mathbf{Hom}}^{\text{dual}}(\mathcal{C}, \mathcal{C})$$

a quotient, so the internal identity lifts to some $u \in \underline{\mathbf{Hom}}^{\text{dual}}(\mathcal{C}, \mathcal{P})$ with $q \circ u \simeq \mathbf{1}_{\mathcal{C}}$. For a family $(\mathcal{D}_s)$ of dualizable categories, write

$$\Theta_{\mathcal{X}} : \left(\prod_s^{\text{dual}} \mathcal{D}_s \right) \otimes \mathcal{X} \longrightarrow \prod_s^{\text{dual}} (\mathcal{D}_s \otimes \mathcal{X}).$$

Functoriality of parameterized evaluation gives $\Theta_{\mathcal{C}} \rightarrow \Theta_{\mathcal{P}}$ from u and the reverse map from q ; their composite is the identity. Thus $\Theta_{\mathcal{C}}$ is a retract of $\Theta_{\mathcal{P}}$. Since $\mathcal{P}$ is proper, $\Theta_{\mathcal{P}}$ is fully faithful; therefore so is $\Theta_{\mathcal{C}}$. Efimov's product criterion [2, Proposition 3.18] now implies that $\mathcal{C}$ is proper. $\square$

References

- [1] A. I. Efimov, *K-theory and localizing invariants of large categories*, arXiv:2405.12169v3.
- [2] A. I. Efimov, *Localizing invariants of inverse limits*, arXiv:2502.04123v2.
- [3] J. Lurie, *Kerodon*, Tag 01NC, “The Category of Simplices.”